

Open WebUI

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Users using the AI large model image have already set up their environment. No setup-related steps are required in this tutorial; simply jump to the running section.

Demo Environment

Development Board: Jetson Orin series mainboard

Open WebUI is an open-source project designed to provide a simple and easy-to-use user interface (UI) for managing and monitoring open-source software and services.

when using Open webUI, there's a high probability of encountering unresponsive dialogues or timeouts. Try restarting Open webUI or using the ollam tool to run the model!

1. Environment Requirements

Host machine and Conda installation of Open WebUI: Requires Node.js >= 20.10, Python = 3.11:

Environment Setup Method	Difficulty (relatively speaking)
Host	High
Conda	Medium
Docker	Low

This tutorial demonstrates installing Open WebUI using Docker.

2. Docker Setup

2.1 Official Docker Installation

If Docker is not installed, you can use a script to install it with one click.

`` Our provided mainboard image already has Docker installed; you don't need to install it yourself!

Domestic users may not be able to install Docker through the official methods. We recommend using Alibaba Cloud to install Docker or directly using our image.

- Update local package list

```
sudo apt update
```

- Upgrade installed packages

```
sudo apt upgrade
```

- Download and run script

Download the get-docker.sh file and save it in the current directory.

```
sudo apt install curl
```

```
curl -fsSL https://get.docker.com -o get-docker.sh
```

Run the get-docker.sh script file with sudo privileges.

```
sudo sh get-docker.sh
```

2.2 Installing Docker on Alibaba Cloud

If you cannot install it yourself, please use the image we provide.

- Update local package list

```
sudo apt update
```

- Install essential software

```
sudo apt install apt-transport-https ca-certificates curl gnupg2 lsb-release  
software-properties-common
```

- Add GPG key for software repository
 - ORIN users

```
curl -fsSL https://mirrors.aliyun.com/docker-ce/linux/ubuntu/gpg | sudo gpg  
--dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg
```

- PI5, RDK X5 users

```
curl -fsSL https://mirrors.aliyun.com/docker-ce/linux/debian/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg
```

- Add Alibaba Cloud mirror software source
 - ORIN user

```
echo "deb [arch=arm64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg] https://mirrors.aliyun.com/docker-ce/linux/ubuntu bionic stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

- PI5, RDK X5 user

```
echo "deb [arch=arm64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg] https://mirrors.aliyun.com/docker-ce/linux/debian bookworm stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

- Install Docker

```
sudo apt update  
sudo apt install docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

2.3 Add Access Permissions

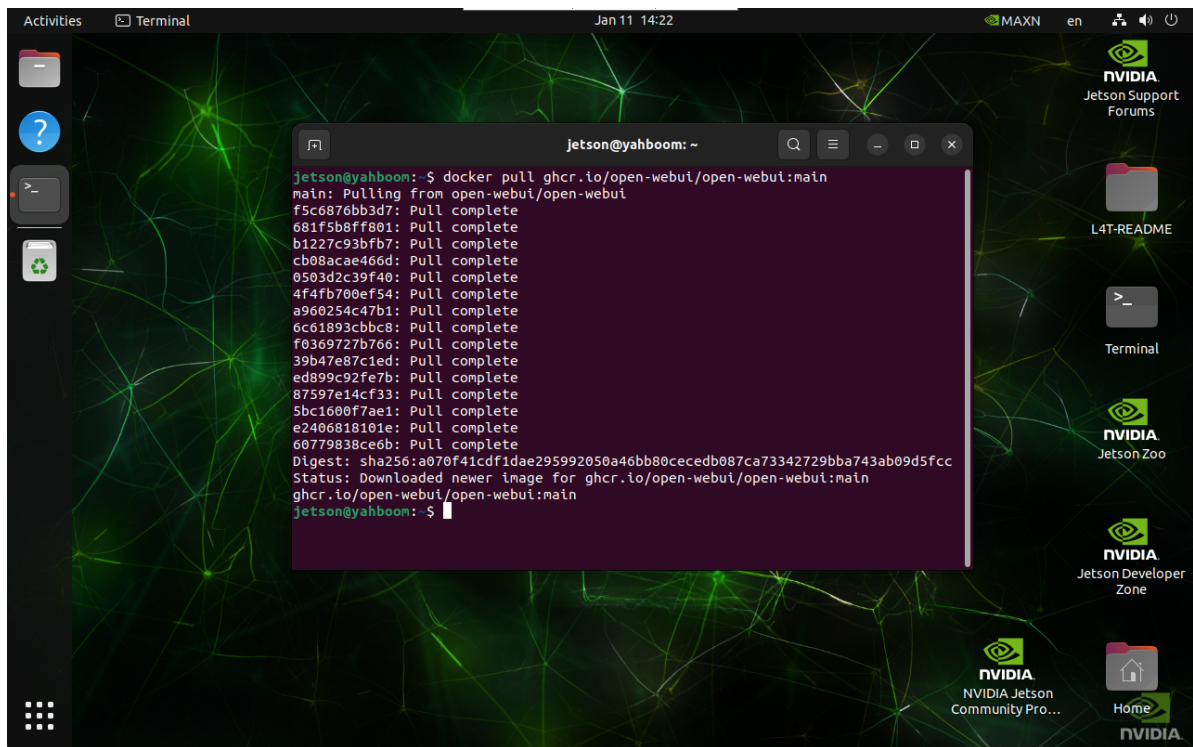
Add access permissions for the current system user to the Docker daemon: Docker commands can be used without using the sudo command.

```
sudo usermod -aG docker $USER  
newgrp docker
```

3. Install Open WebUI

For systems with Docker already installed, you can directly enter the following command in the terminal: The image shows the result after pulling.

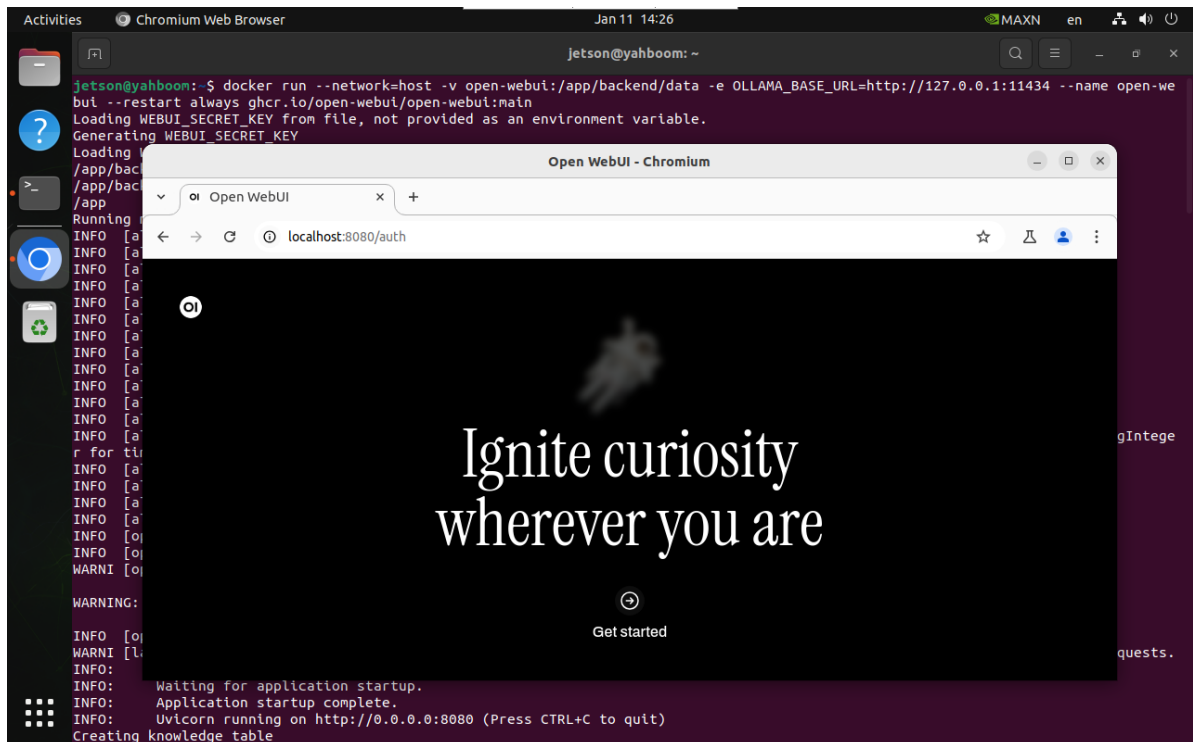
```
docker pull ghcr.io/open-webui/open-webui:main
```



4. Run Open WebUI

Enter the following command in the terminal to start the specified Docker container:

```
docker run --network=host -v open-webui:/app/backend/data -e  
OLLAMA_BASE_URL=http://127.0.0.1:11434 --name open-webui --restart always  
ghcr.io/open-webui/open-webui:main
```



After successful startup, access it in your browser using the following URL:

```
http://localhost:8080/
```

On the same local area network, you can access it using the mainboard IP:8080:

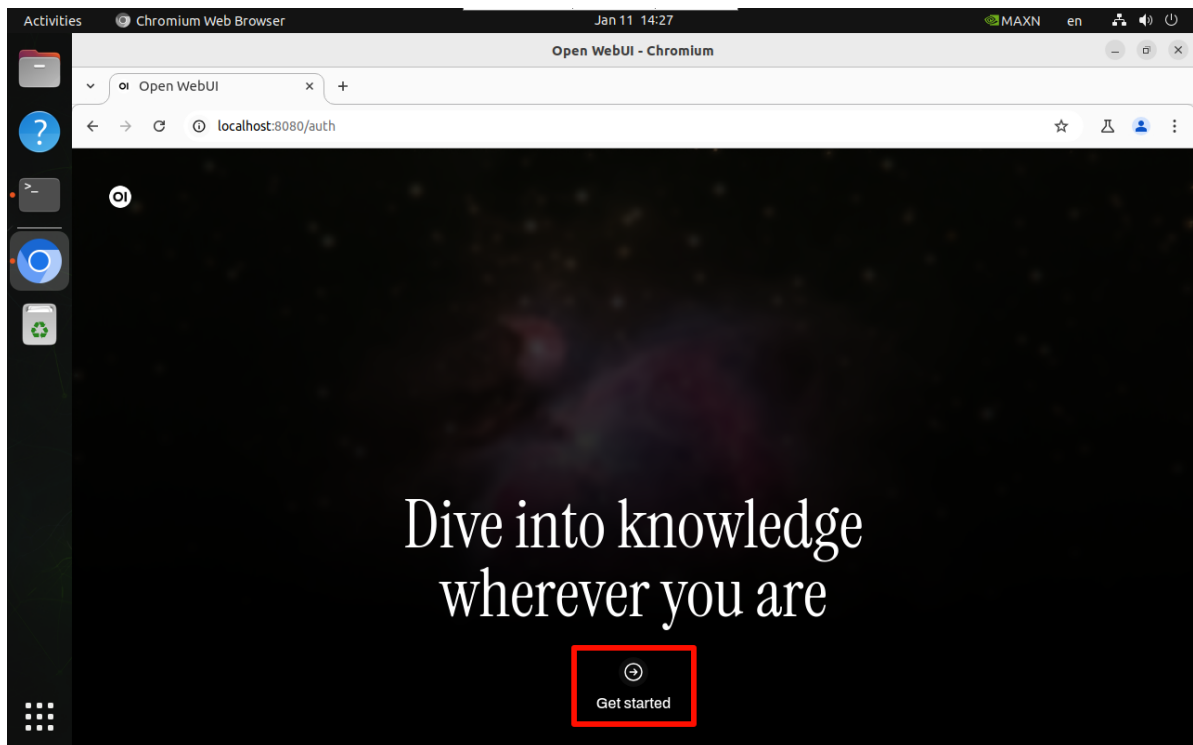
Assuming the mainboard IP: 192.168.2.105, we can access it via 192.168.2.105:8080

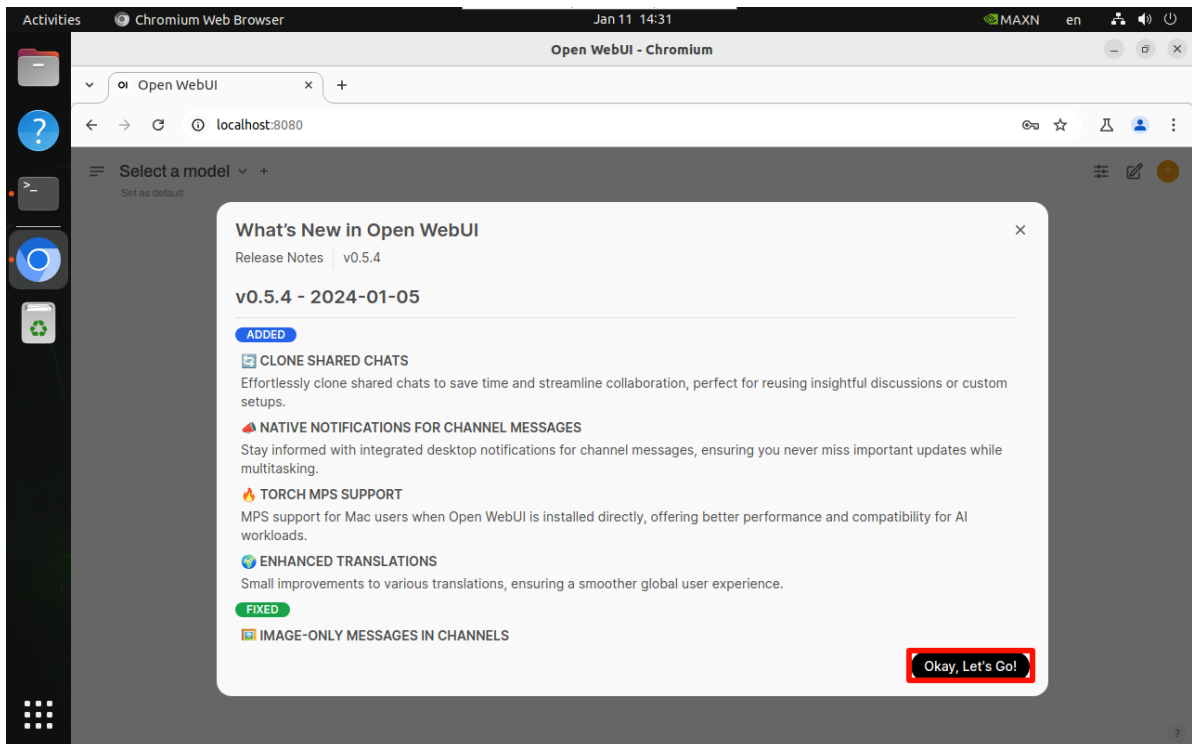
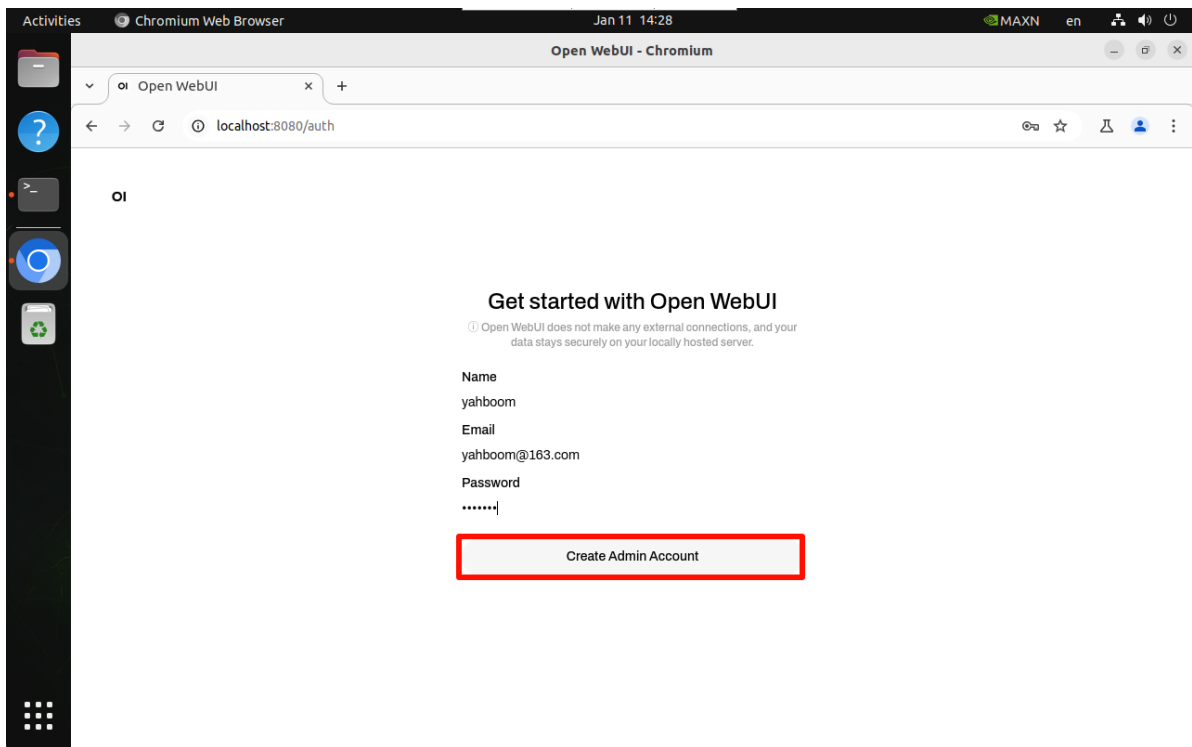
4.1 Administrator Account

First-time users need to register an account. This account is the administrator account. Fill in the required information as instructed!

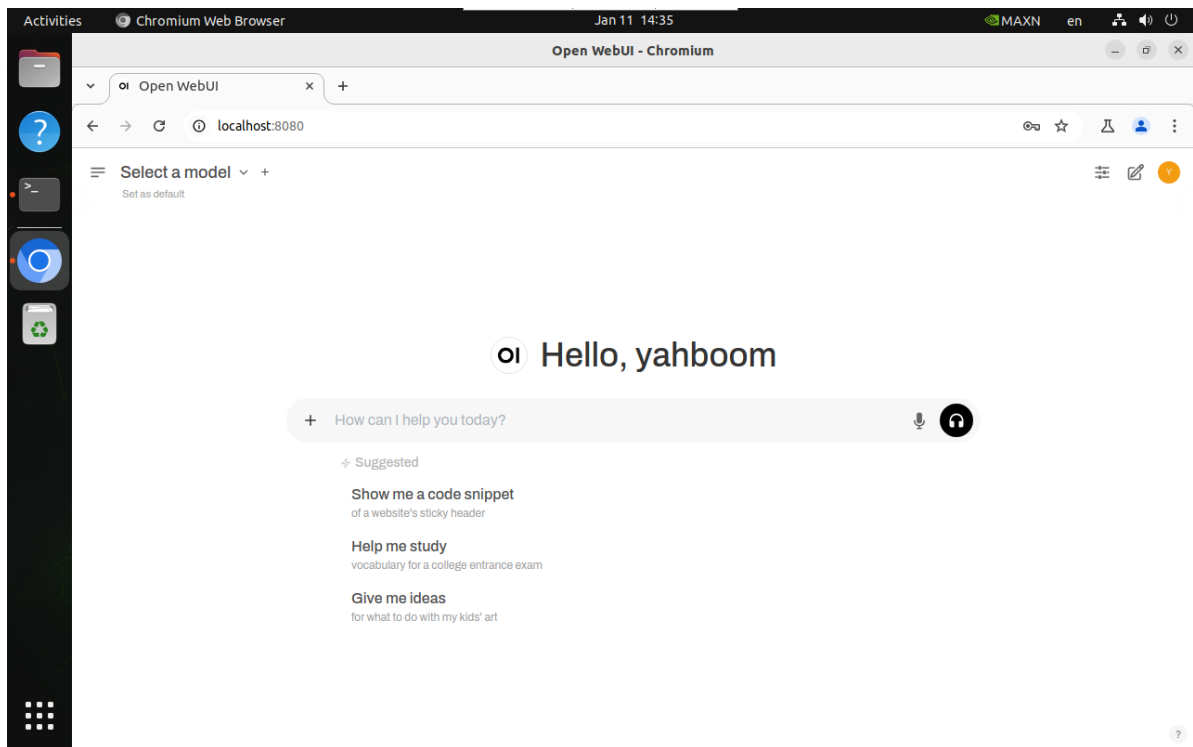
Since all content in our mirror has been set up and tested, users can directly log in using their registered account:
Username: yahboom
Email: yahboom@163.com
Password: yahboom

4.2 Registration and Login





4.3 User Interface



5. Model Dialogue

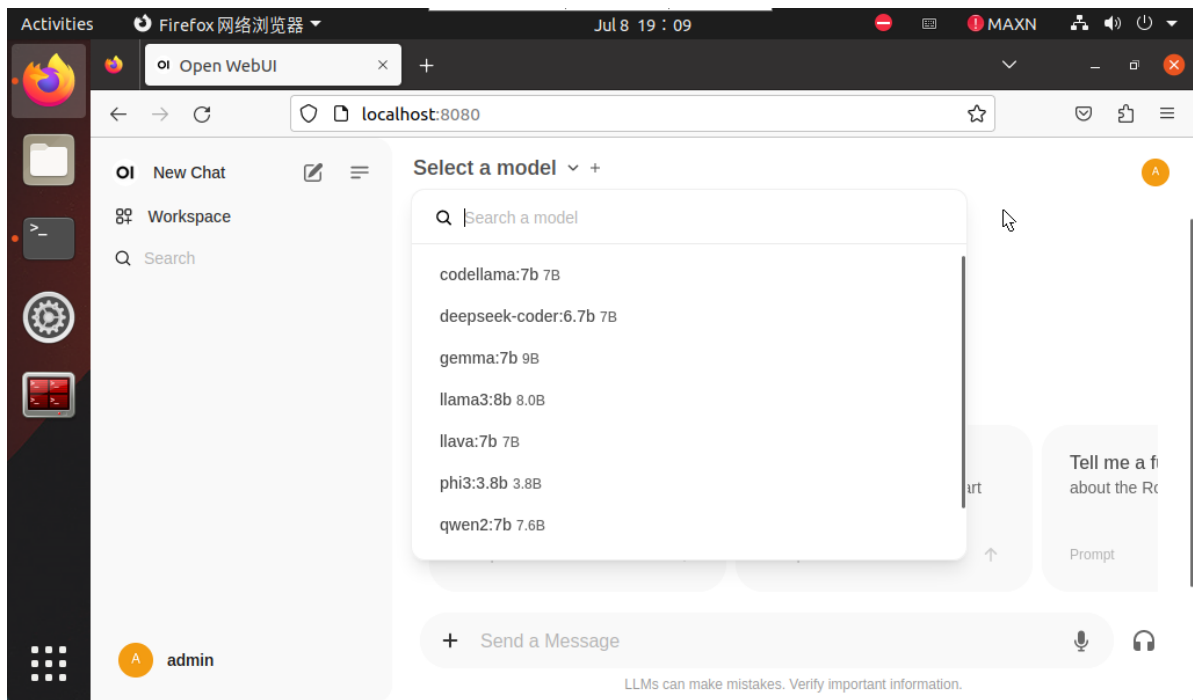
Using the Open WebUI for dialogue is slower than using the Ollam tool directly, and may even result in timeouts and connection failures. This is related to the mainboard's memory and is unavoidable!

Users interested can set up Ollam and Open WebUI on other Linux environments for dialogue.

5.1 Switching Models

Clicking `Select a model` allows you to select a specific model for dialogue.

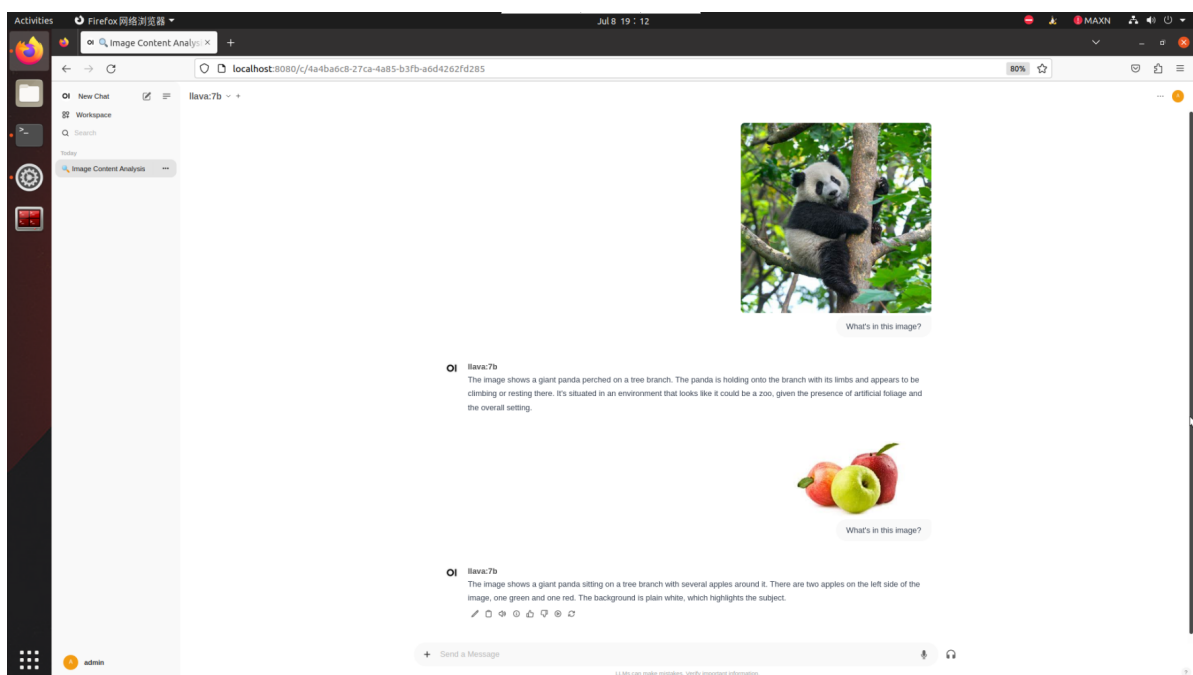
Models retrieved using ollam are automatically added to the open webUI model options. Refreshing the webpage will display the new model!



5.2 Demo: LLaVA

The LLaVA demo requires 8GB or more to run. Users can use other demos to test the Open WebUI conversational functionality!

what's in this image?



6. Frequently Asked Questions

6.1 Disabling Open WebUI

Disable the automatic startup of Open WebUI.

- View running Docker containers

```
docker ps
```


- Shut down running Docker containers

```
docker stop [CONTAINER ID] # Example docker stop 5f42ee9cf784
```

- View stopped containers

```
docker ps -a
```

- Clean up stopped containers

```
docker rm [CONTAINER ID] # Example docker rm 5f42ee9cf784
```

Clean up all stopped containers:

```
docker container prune
```

6.2 Common Errors

Unable to start Open WebUI

- docker: Error response from daemon: Conflict. The container name "/open-webui" is already in use by container "cfc05c84f8e38b290337e7178c76fd1c49076f94b11ed3d49d9448be72b7f20f". You have to remove (or rename) that container to reuse that name.

Solution: Close and restart Open WebUI!

Service Connection Timeout

- Open WebUI: Server Connection Error

Close and restart Open WebUI, then try asking the question again, or use the Ollam tool to run the model and ask the question!